Description

Image





Caption

1. Moka Express Espresso Maker made of aluminum alloy, cast in three parts (made by Bialetti). © Hans Chr. R. at en.wikipedia - (CC BY-SA 3.0) 2. Aluminum casting alloys almost all contain silicon to make them fluid, allowing castings with good finish and detail.

The material

Almost all aluminum alloys for casting contain 5 - 22% silicon (Si) -- the silicon makes the alloys more fluid so that they fill the mold and take up fine detail, even in thin sections. Further additions of copper (Cu) or magnesium (Mg) give age-hardening alloys. The plain Al-Si alloys are used for marine components and hardware and for cooking utensils because of their good resistance to corrosion in salt water; and they are used for pistons and cylinder liners because of their good thermal conductivity and low expansion. As a general rule the casting alloys have lower ductility and strength than the wrought age-hardening alloys -- few have tensile strengths above 350 MPa.
An alternative name for Aluminum in many countries is Aluminium.

Compositional summary

Al + 5 - 22% Si, sometimes with some Cu, Mg or Zn to allow age-hardening.

General properties

Density		2.65e3	-	2.77e3	kg/m^3
Price	*	1.98	-	2.1	GBP/kg
Date first used		1905			

Mechanical properties

Young's modulus		69	-	76	GPa
Shear modulus	*	26	-	29	GPa
Bulk modulus	*	66	-	76	GPa
Poisson's ratio		0.32	-	0.34	

Yield strength (elastic limit)		118	-	263	MPa
Tensile strength		193	-	341	MPa
Compressive strength	*	109	-	251	MPa
Elongation		1.2	-	6.8	% strain
Hardness - Vickers		83	-	116	HV
Fatigue strength at 10^7 cycles		63	-	136	MPa
Fracture toughness	*	19	-	30.9	MPa.m^0.5
Mechanical loss coefficient (tan delta)	*	0.003	-	0.044	

Thermal properties

Melting point		535	-	639	°C
Maximum service temperature		138	-	200	°C
Minimum service temperature		-273			°C
Thermal conductor or insulator?		Good conductor			
Thermal conductivity		110	-	162	W/m.°C
Specific heat capacity		944	-	982	J/kg.°C
Thermal expansion coefficient		19.5	-	23.3	µstrain/°C

Electrical properties

Electrical conductor or insulator?		Good conductor			
Electrical resistivity		4	-	6.5	µohm.cm

Optical properties

Transparency	Opaque

Critical Materials Risk

High critical material risk?	Yes
Notes	Al (aluminum) added to the 2018 US critical minerals list

Processability

Castability	4	-	5	
Formability	3	-	4	
Machinability	4	-	5	
Weldability	3	-	4	
Solder/brazability	2	-	3	

Durability: water and aqueous solutions

Water (fresh)	Excellent
Water (salt)	Acceptable
Soils, acidic (peat)	Unacceptable

Soils, alkaline (clay)	Excellent
Wine	Excellent

Durability: acids

Acetic acid (10%)	Limited use
Acetic acid (glacial)	Unacceptable
Citric acid (10%)	Acceptable
Hydrochloric acid (10%)	Limited use
Hydrochloric acid (36%)	Unacceptable
Hydrofluoric acid (40%)	Unacceptable
Nitric acid (10%)	Limited use
Nitric acid (70%)	Acceptable
Phosphoric acid (10%)	Unacceptable
Phosphoric acid (85%)	Unacceptable
Sulfuric acid (10%)	Limited use
Sulfuric acid (70%)	Unacceptable

Durability: alkalis

Sodium hydroxide (10%)	Unacceptable
Sodium hydroxide (60%)	Unacceptable

Durability: fuels, oils and solvents

Amyl acetate	Excellent
Benzene	Excellent
Carbon tetrachloride	Excellent
Chloroform	Excellent
Crude oil	Excellent
Diesel oil	Excellent
Lubricating oil	Excellent
Paraffin oil (kerosene)	Excellent
Petrol (gasoline)	Excellent
Silicone fluids	Excellent
Toluene	Excellent
Turpentine	Excellent
Vegetable oils (general)	Excellent
White spirit	Excellent

Durability: alcohols, aldehydes, ketones

Acetaldehyde	Excellent

Acetone	Excellent
Ethyl alcohol (ethanol)	Acceptable
Ethylene glycol	Excellent
Formaldehyde (40%)	Excellent
Glycerol	Limited use
Methyl alcohol (methanol)	Acceptable

Durability: halogens and gases

Chlorine gas (dry)	Limited use
Fluorine (gas)	Unacceptable
O2 (oxygen gas)	Excellent
Sulfur dioxide (gas)	Acceptable

Durability: built environments

Industrial atmosphere	Excellent
Rural atmosphere	Excellent
Marine atmosphere	Excellent
UV radiation (sunlight)	Excellent

Durability: flammability

Flammability	Non-flammable

Durability: thermal environments

Tolerance to cryogenic temperatures	Excellent
Tolerance up to 150 C (302 F)	Acceptable
Tolerance up to 250 C (482 F)	Unacceptable
Tolerance up to 450 C (842 F)	Unacceptable
Tolerance up to 850 C (1562 F)	Unacceptable
Tolerance above 850 C (1562 F)	Unacceptable

Geo-economic data for principal component

Annual world production, principal component	5.73e7	tonne/yr
Reserves, principal component	2.8e10	tonne

Primary material production: energy, climate change and water

Climate change (CO2-eq) (virgin grade production)	* 12	-	13.3	kg/kg
Embodied energy (virgin grade production)	* 165	-	182	MJ/kg
Water usage	* 1.05e3	-	1.16e3	l/kg

Material processing: energy

Metal casting, embodied energy	* 48.7	-	55.7	MJ/kg
Chemical vapor deposition (CVD), embodied energy	* 1.42e4	-	1.61e4	MJ/kg
Machining, coarse, embodied energy	* 17	-	19	MJ/kg
Machining, fine, embodied energy	* 21.8	-	24.7	MJ/kg
Machining, grinding, embodied energy	* 26.6	-	30.4	MJ/kg
Machining, laser, CO2, embodied energy	* 1.69	-	1.92	MJ/m

Material processing: climate change

Metal casting, climate change (CO2-eq)	* 3.52	-	4.04	kg/kg
Chemical vapor deposition (CVD), climate change (CO2-eq)	* 1.17e3	-	1.32e3	kg/kg
Machining, coarse, climate change (CO2-eq)	* 1.11	-	1.24	kg/kg
Machining, fine, climate change (CO2-eq)	* 1.43	-	1.62	kg/kg
Machining, grinding, climate change (CO2-eq)	* 1.74	-	1.99	kg/kg
Machining, laser, CO2, climate change (CO2-eq)	* 0.139	-	0.158	kg/m

Recycling and end of life

Recycle	✓			
Functional recycle	✗			
Climate change (CO2-eq), recycling	* 2.76	-	3.06	kg/kg
Embodied energy, recycling	* 47.8	-	52.8	MJ/kg
Recycle fraction in current supply	33			%
Combust for energy recovery	✗			
Landfill	✓			
Biodegrade	✗			
Toxicity rating	Non-toxic			
A renewable resource?	✗			

Environmental notes

Aluminum ore is abundant. It takes a lot of energy to extract aluminum, but it is easily recycled at low energy cost.

Supporting information

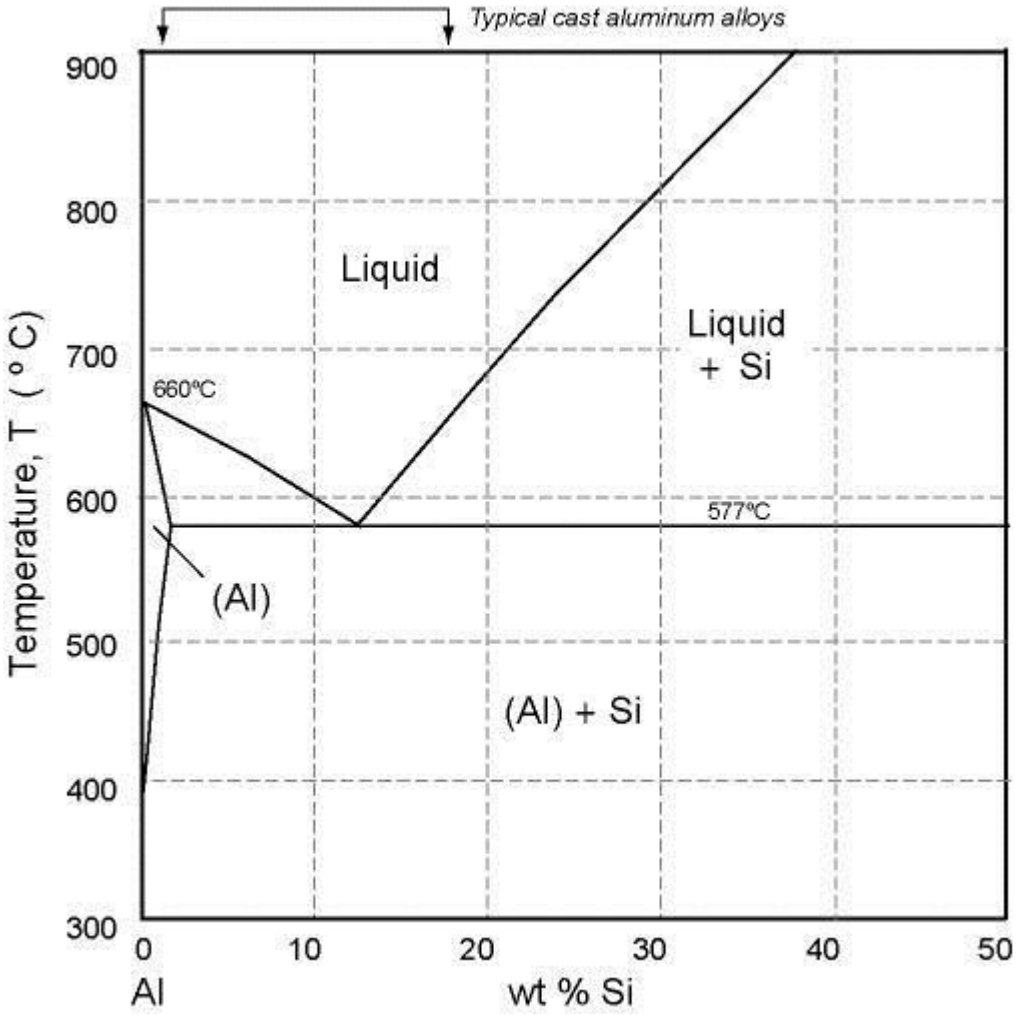
Design guidelines

Aluminum casting alloys are designed to be fluid so that they fill the mold and take up details well -- alloying with silicon is the most effective way to do this. Some of the alloys are designed for die-casting, some for permanent mold casting and some for sand casting. The main challenge with aluminum casting is coping with the shrinkage of 3.5 to 8.5 % that happens during solidification; this requires mold design to get the right final dimensions and to avoid hot tearing, cracking or porosity. Despite this constraint, aluminum castings of great complexity are practical. Recent developments in rheo-casting and squeeze casting overcome some of the problems of dimensional accuracy. The mechanical properties of the cast alloys are less good and more variable than those of the wrought series.

Technical notes

No classification system for cast aluminum alloys has international acceptance. In the most widely used (the AAUS system), the first digit indicates the alloy group. In the 1xx.x group, the second two digits indicate the minimum percentage of aluminum; thus 150.x indicates a composition containing a minimum of 99.5% aluminum. The digit to the right of the decimal point indicates the product form: 0 means 'castings' and 1 means 'ingot'. In the 2xx.x to 9xx.x groups, the second two digits are simply serial numbers. The digit to the right of the decimal point again indicates product form. Age-hardening alloys carry the suffix T and a number between 0 and 8 to indicated the state of heat treatment.

Phase diagram



Phase diagram description

Most cast aluminum alloys are based on alloys of aluminum (Al) with 1 - 18% silicon (Si), for which this is the phase diagram.

Typical uses

Aerospace engineering, automotive engineering - pistons, clutch housings, exhaust manifolds, die cast chassis and casings for household and electronic products.

Links

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